

A disproof of conjecture 00000001202

Octal game 0.07 is not eventually $(12, 4)$ -periodic

Abstract

Conjecture 00000001202 asserts that the Sprague–Grundy sequence of the octal game 0.07 is eventually periodic with period 12 and pre-period length 4, i.e. that $g(n + 12) = g(n)$ for all $n \geq 4$. We show the conjecture is *false* under both natural readings of the definition. Under the standard octal reading ($d_1 = 0$, $d_2 = 7$; a move removes exactly two tokens) the game is Dawson’s Kayles and the identity already fails at $n = 4$: $g(4) = 2$ whereas $g(16) = 5$. The true eventual period is 34 with pre-period 53. Under the conjecture’s own prose reading (“a move removes 1 or 2 tokens”, i.e. octal 0.77, Kayles) the period 12 is correct but the pre-period is 71, not 4; the identity fails first at $n = 10$, where $g(10) = 2$ but $g(22) = 6$. In both cases the asserted universality from $n \geq 4$ is refuted. The counterexample is formalised in core Lean 4 and reproduced by a standalone Python script.

1 Octal games and Sprague–Grundy values

An *octal game* is played on heaps of tokens. A move chooses a heap of size n , removes exactly $k \geq 1$ tokens from it, and splits the resulting $n - k$ tokens into some number of heaps. The game is encoded by a sequence of octal digits $d_1 d_2 d_3 \dots$, where the binary expansion of d_k records which splits of the remainder are legal (bit j corresponds to splitting the remainder into j heaps). The Sprague–Grundy (SG) value $g(n)$ of a single heap of size n is

$$g(n) = \text{mex}\{g(a) \oplus g(b) : \text{legal move to heaps } a, b\}, \quad (1)$$

where $\oplus$ is the bitwise XOR (nim-sum), and $\text{mex } S = \min(\mathbb{N} \setminus S)$.

The conjecture fixes the game “0.07” and offers two prose descriptions, which unfortunately determine different games.

1.1 Reading A: standard octal 0.07 (Dawson’s Kayles)

Here $d_1 = 0$ and $d_2 = 7 = 111_2$. Thus *no* move may remove a single token, while a move may remove exactly two tokens and may split the remaining $n - 2$ tokens into 0, 1, or 2 heaps (all three splits are legal). Taking $a, b \geq 0$ as the resulting heap sizes, (1) becomes, for $n \geq 2$,

$$\boxed{g(n) = \text{mex}\{g(a) \oplus g(b) : a + b = n - 2\}}, \quad (2)$$

with no legal move for $n < 2$ (so $g(0) = g(1) = 0$). This is *Dawson’s Kayles*.

1.2 Reading B: the prose “removes 1 or 2 tokens” (octal 0.77)

If one instead allows removal of either one or two tokens, both with all three splits of the remainder, the game is 0.77 ($d_1 = d_2 = 7$), i.e. *Kayles*, and (1) becomes

$$g(n) = \text{mex}\left(\{g(a) \oplus g(b) : a + b = n - 1\} \cup \{g(a) \oplus g(b) : a + b = n - 2\}\right). \quad (3)$$

So 0.07 is *not* literally “remove 1 or 2 tokens”: the two descriptions in the conjecture are inconsistent.

2 The conjectured statement

The conjecture claims

$$g(n + 12) = g(n) \quad \text{for all } n \geq 4, \quad (4)$$

i.e. eventual period 12 with pre-period length 4 (the values $g(0), \dots, g(3)$ being the exceptional pre-period). We refute (4) under both readings.

3 Reading A: the identity fails at $n = 4$

Iterating (2) gives the following initial values.

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$g(n)$	0	0	1	1	2	0	3	1	1	0	3	3	2	2	4	0	5	2

In particular

$$g(4) = 2, \quad g(16) = 5.$$

Since $16 = 4 + 12$, the conjectured identity (4) would force $g(16) = g(4)$; instead $g(16) = 5 \neq 2 = g(4)$. And $n = 4$ is the *first* index at which (4) fails, so the failure is as early as possible.

Theorem 1 (Refutation at $n = 4$). *Under reading A, $g(4) \neq g(16)$; indeed $g(4) = 2$ and $g(16) = 5$. Consequently (4) is false.*

Proof. Direct evaluation of (2). The reachable nim-sums for $n = 4$ are $g(0) \oplus g(2) = 0 \oplus 1 = 1$, $g(1) \oplus g(1) = 0$, and $g(2) \oplus g(0) = 1$, so $\text{mex}\{0, 1\} = 2$. For $n = 16$ the values are $g(0..14) = (0, 0, 1, 1, 2, 0, 3, 1, 1, 0, 3, 3, 2, 2, 4)$; the reachable set is $\{0, 1, 2, 3, 4\}$ and hence $g(16) = \text{mex}\{0, 1, 2, 3, 4\} = 5$. These computations are reproduced exactly by the Lean formalisation (Theorem `period12_fails_at_4` in `lean4/Main.lean`) and by `reproduce.py`. $\square$

4 Reading A: the true eventual period is 34, pre-period 53

Exhaustive computation of (2) shows that the sequence eventually has period 34, and 53 is the smallest index from which that period holds. That is,

$$g(n + 34) = g(n) \quad \text{for all } n \geq 53,$$

and no smaller period works from any smaller index. In particular the conjectured “period 12, pre-period 4” is wrong on both counts (the statement’s own table of values already contradicts it at $n = 4$). We verified this to $n = 1000$ with `reproduce.py`; the value $g(53 + d)$ is confirmed to agree with $g(53 + 34 + d)$ throughout the computed range $d = 0, \dots, 913$.

object	value
claimed period	12
claimed pre-period	4
true eventual period of 0.07	34
true pre-period of 0.07	53
first failure of $g(n + 12) = g(n)$ for $n \geq 4$	$n = 4$
witness	$g(4) = 2 \neq 5 = g(16)$

5 Reading B: period 12 is right, but the pre-period is 71

For the prose reading, i.e. octal 0.77 and recurrence (3), the initial values are

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
$g(n)$	0	1	2	3	1	4	3	2	1	4	2	6	4	1	2	7	1	4	3	2	1	4	6	7

Here the eventual period really is 12, but the pre-period is 71, not 4. Consequently (4) still fails: the first index $n \geq 4$ with $g(n + 12) \neq g(n)$ is

$$n = 10, \quad g(10) = 2, \quad g(22) = 6.$$

Thus even granting the prose reading, the claimed universality from $n \geq 4$ is false.

object	value
claimed pre-period	4
true eventual period of 0.77	12
true pre-period of 0.77	71
first failure of $g(n + 12) = g(n)$ for $n \geq 4$	$n = 10$
witness	$g(10) = 2 \neq 6 = g(22)$

6 Numerical and formal verification

Numerical. `reproduce.py` (standard library only) implements both recurrences (2) and (3), computes $g(n)$ for $0 \leq n \leq 1000$ (≥ 800 , as required), finds the first $n \geq 4$ violating $g(n+12) = g(n)$, and searches for the true eventual period and minimal pre-period with a confirmation window of hundreds of terms. It prints value tables and PASS/FAIL lines and exits with status 0. For 0.07 it reports first failure $n = 4$ (2 vs. 5), period 34, pre-period 53; for 0.77 it reports first failure $n = 10$ (2 vs. 6), period 12, pre-period 71.

Formal. The Lean 4 development in `lean4/` uses core Lean only (`import Std`, no `Mathlib`, no `sorry`, no `axiom`, no `native_decide`). It defines the fueled `mex` and the computable recurrence (2) by structural recursion, then proves by kernel evaluation

$$g(4) = 2, \quad g(16) = 5, \quad g(4) \neq g(16),$$

and packages them as `conjecture_00000001202_false`. The eventual period 34 / pre-period 53 (and the reading-B data) are established numerically, not in Lean; the formalisation deliberately covers the concrete refuting instance $n = 4$.

7 Conclusion

Conjecture 00000001202 is **false**. Under the standard reading of octal 0.07 (Dawson’s Kayles) the claimed identity $g(n + 12) = g(n)$ for all $n \geq 4$ fails immediately at $n = 4$, where $g(4) = 2$ but $g(16) = 5$; the true eventual period and pre-period are 34 and 53. Under the conjecture’s own prose reading (octal 0.77, Kayles) the period 12 is correct but the pre-period is 71, so the identity still fails (first at $n = 10$, $g(10) = 2$ vs. $g(22) = 6$). The counterexample is reproduced by `reproduce.py` and formalised in `lean4/`.